

Computational Haemodynamics

From a clinical question to a computed answer, and what that answer is worth

From Physics to Physiology · Lecture 3

Scope of the lecture

Part 1 · Aortic coarctation

a clinical question, and the complete image-based CFD workflow that answers it

Part 2 · Coronary stenosis

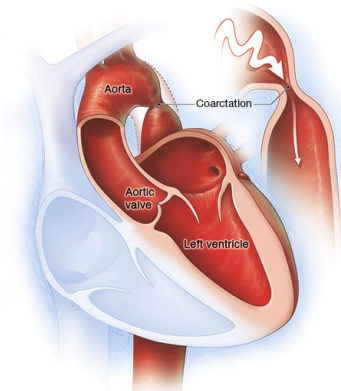
the same workflow, applied to fractional flow reserve from CT (FFR-CT)

One workflow: clinical question → physics → image → geometry → mesh → boundary conditions → solver → validation. Each step introduces an assumption; the computed number inherits all of them.

Part 1 · Aortic coarctation

the condition · the invasive criterion and its limits · computing the pressure drop *in silico* · the image-based CFD workflow, step by step

Aortic coarctation: the condition



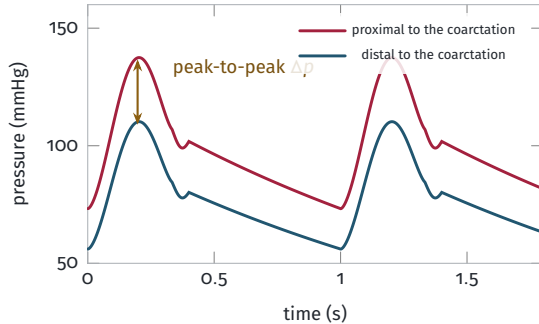
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Coarctation of the aorta: the narrowing sits just distal to the left subclavian artery. Illustration: Mayo Foundation for Medical Education and Research.

- Congenital narrowing of the proximal descending aorta, typically just distal to the left subclavian artery; 5–8% of congenital heart disease
- Consequences: increased left-ventricular afterload, upper-body hypertension, collateral circulation; untreated, reduced life expectancy
- Treatment: surgical repair or balloon angioplasty and stenting; recoarctation is common and requires surveillance

The clinical decision at every stage is a decision about **severity**: is the narrowing haemodynamically significant?

The diagnostic criterion: the invasive pressure drop



- Cardiac catheterisation: a pressure catheter is withdrawn across the narrowing
- Severity index: **peak-to-peak systolic pressure difference**
- Guideline threshold for intervention: $\Delta p > 20 \text{ mmHg}$ (ACC/AHA; ESC)

The reference standard is a **measured** quantity: two pressure traces, one number.

Limitations of the invasive measurement

invasive

arterial access;
general anaesthesia
in children; vascular
complications

exposure

ionising radiation
and iodinated
contrast, repeated
at each follow-up

cost and access

catheterisation
laboratory, staff,
hospital stay

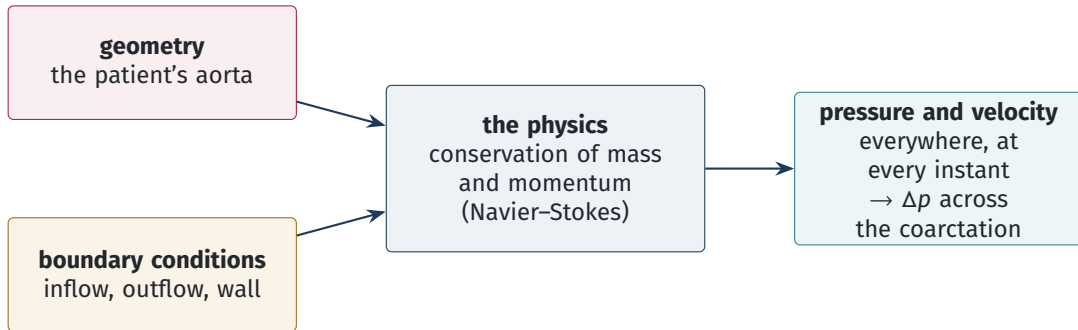
perturbation

the catheter itself
obstructs the nar-
rowing; pressure only
at the catheter tip

- Non-invasive surrogate: Doppler peak velocity and the simplified Bernoulli relation, $\Delta p \approx 4 v_{\max}^2$ (mmHg, m/s), which neglects viscous losses, unsteadiness and pressure recovery

Can the pressure drop be obtained *in silico*, from a non-invasive image, with the full physics retained?

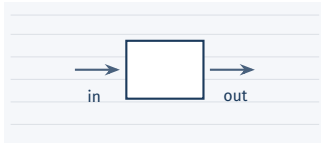
Computing the pressure drop: what would be needed



- Pressure is not measured but **inferred**: the field that, with velocity, satisfies the conservation laws in this geometry with these boundary conditions
- This is computational fluid dynamics (CFD): the numerical solution of the governing equations in a patient-specific domain

The physics: the Navier–Stokes equations

a control volume fixed in space



1 • Conservation of mass

net inflow equals net outflow (incompressible)

2 • Conservation of momentum

rate of change of momentum

= pressure force + viscous force

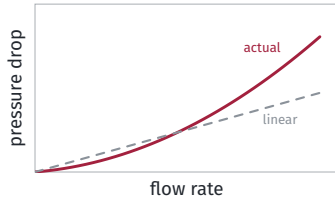
BOARD WORK

$$\underbrace{\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right)}_{\text{inertia: local + convective}} = \underbrace{-\nabla p}_{\text{pressure gradient}} + \underbrace{\mu \nabla^2 \mathbf{u}}_{\text{viscous diffusion}} \quad \underbrace{\nabla \cdot \mathbf{u} = 0}_{\text{incompressibility}}$$

Four unknowns, four equations; blood as an incompressible Newtonian continuum, $\rho = 1060 \text{ kg/m}^3$, $\mu \approx 3.5\text{--}4 \text{ mPa s}$.

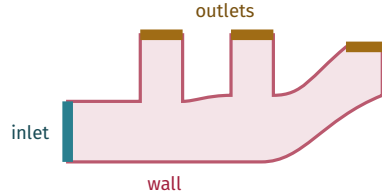
Why the equations cannot be solved analytically

(a) nonlinearity



the convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$: solutions cannot be superposed

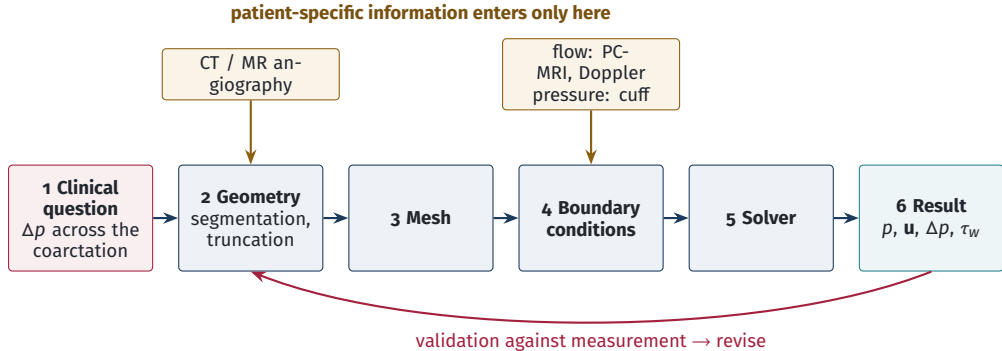
(b) the solution depends on domain and boundaries



closed-form solutions exist only for idealised cases (Poiseuille, Womersley); an irregular geometry and pulsatile boundary data admit none

Pressure has no equation of its own: it enforces incompressibility. The solution must be **approximated numerically**, on a discretised domain, with prescribed boundary conditions.

The image-based CFD workflow



The governing equations are identical for every patient. This diagram is the outline of the rest of Part 1.

Step 2 · From image to geometry: segmentation

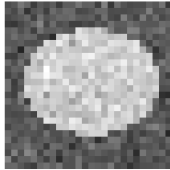
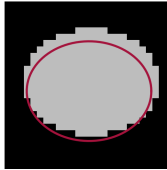
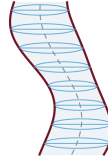


image (CT / MR): intensities,
partial-volume blur at edges



segmentation: binary label map
+ extracted lumen contour

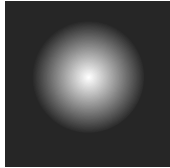


stacked contours → 3-D surface,
centerline and radius $r(s)$

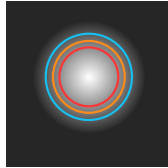
- The image is a grid of intensities; the lumen boundary is not in the data and must be **extracted**: thresholding, region growing, or a trained neural network
- The result is a binary label map, then a closed triangulated surface: the computational domain

Sources: CT angiography (0.3–0.6 mm, contrast and radiation) or MR angiography (0.8–1.5 mm, no radiation).

Segmentation is a modelling decision, not a measurement



acquired data: partial-volume blur



three defensible thresholds



three different domains

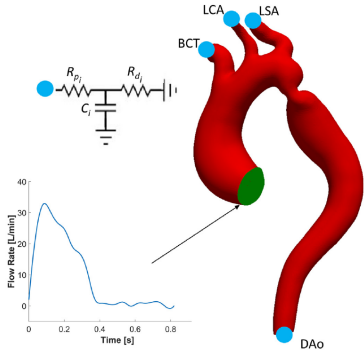
BOARD WORK

Poiseuille, fixed Q : $\Delta p \propto R^{-4}$, $\tau_W \propto R^{-3}$; R under-read by 10%: $(1/0.9)^4 = 1.52$, i.e. +52% in Δp

Inter-observer variability in radius is 5–10%; in a 3–4 mm coarctation this propagates to tens of per cent in Δp . The lumen boundary is the first-order determinant of accuracy.

Modelling choice I · Where to truncate the domain

b)

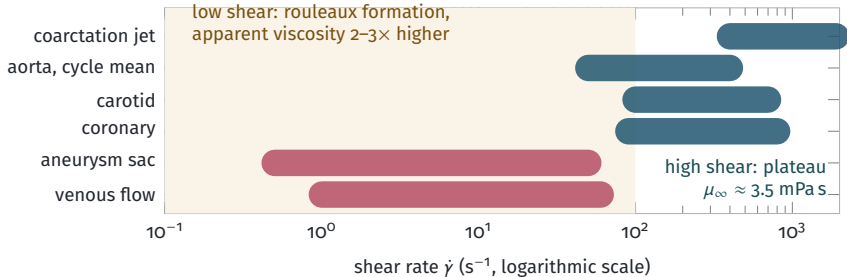


Computational domain of a coarctation patient: inlet at the ascending aorta, outlets at the three arch branches and the descending aorta.

- The model cannot contain the whole circulation; every truncation plane becomes a boundary requiring a condition
- Retain branches carrying more than a few per cent of the inflow; lump the remainder into the parent outlet
- Truncate perpendicular to the centreline, in straight segments away from the lesion; add cylindrical extensions of 3–5 diameters

An omitted branch is flow silently redistributed among the branches retained.

Modelling choice II · Rheology: is a constant viscosity adequate?



- Blood is shear-thinning, but above $\approx 100 \text{ s}^{-1}$ its viscosity is nearly constant; in a coarctation jet the shear rate is far higher
- Newtonian, $\mu = 3.5\text{--}4 \text{ mPa s}$, is adequate here; a shear-thinning model (e.g. Carreau) is warranted in aneurysm sacs, recirculation zones and veins

Modelling choice III · Flow regime: Reynolds and Womersley numbers

BOARD WORK

$$\text{Re} = \frac{\rho U D}{\mu} \quad \alpha = R \sqrt{\frac{\omega \rho}{\mu}}, \quad \delta = \sqrt{\frac{2\mu}{\rho \omega}}$$

ascending aorta, $D = 28$ mm: $\text{Re} \approx 1700$ (mean), 5500 (peak); $\alpha \approx 20$; $\delta \approx 1$ mm

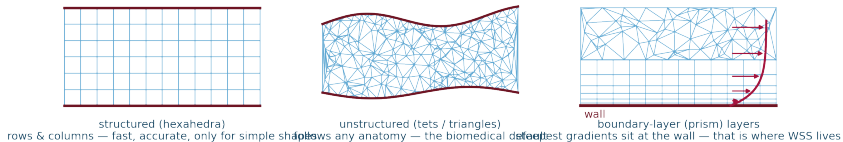
coarctation, $D \approx 10$ mm, $U \approx 2$ m/s:

$\text{Re} \approx 6000$ at peak

- **Re**: inertia against viscosity. Peak values exceed 2300, yet the flow is accelerating, brief and in a short curved vessel: **transitional**, not developed turbulence
- α : pulsatile inertia against viscous diffusion. $\alpha \approx 20$: plug-like core, viscous effects confined to a **Stokes layer of ≈ 1 mm**

Misconception. “ $\text{Re} > 2300$, hence a turbulence model is required.” The threshold applies to steady developed pipe flow; RANS closures add an artificial viscosity and fail silently. Standard practice: no closure, adequate resolution.

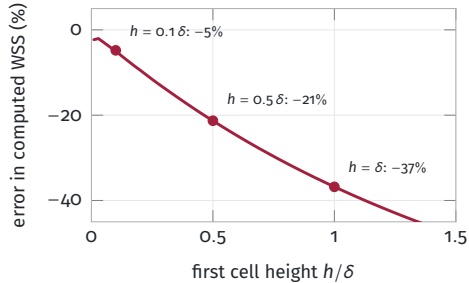
Step 3 · Meshing: why the domain must be discretised



- The equations cannot be solved on an arbitrary shape: the domain is subdivided into cells, and conservation is enforced **cell by cell**
- Tetrahedra conform to any anatomy; prisms form thin layers at the wall; an aortic model has $\approx 4\text{--}6$ million elements

Finer cells: a discrete solution closer to the continuous one, and a longer computation. The error decreases with cell size; it never vanishes.

The near-wall mesh and the wall shear stress



BOARD WORK

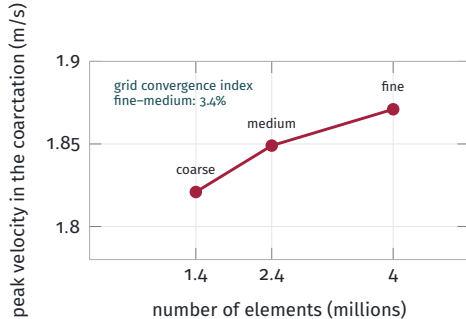
near-wall profile: $u(y) = U(1 - e^{-y/\delta})$; exact $\tau_w = \mu \partial u / \partial y|_0 = \mu U / \delta$

first cell of height h : $\tau_h \approx \mu u(h)/h = (\mu U/h)(1 - e^{-h/\delta})$

$\tau_h/\tau_w = (\delta/h)(1 - e^{-h/\delta}) \rightarrow \delta = 1 \text{ mm}: h \lesssim 0.1 \text{ mm}$

WSS is evaluated from the first cell, which must be a small fraction of the Stokes layer. A mesh without boundary layers yields a converged, plausible and wrong WSS.

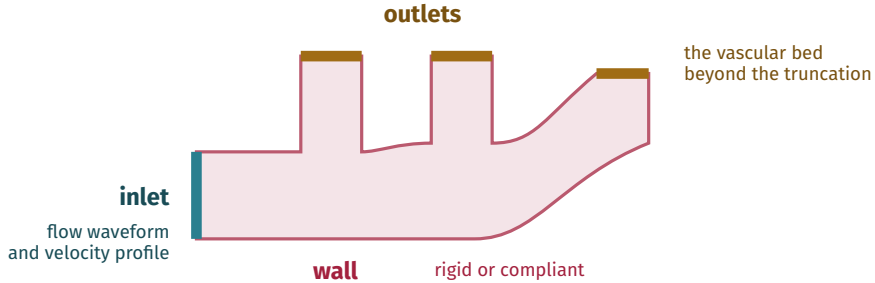
Mesh independence



- Three meshes at a constant refinement ratio; Richardson extrapolation; grid convergence index (GCI)
- Assess convergence on the **reported quantity**: here the peak velocity in the narrowing
- Halving the element size costs ≈ 16 times in a transient run

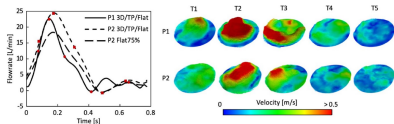
“Refined until it stopped changing” is not a mesh study; a quantitative band on the reported quantity is.

Step 4 · Boundary conditions: where the patient enters



- The equations are patient-independent; specificity resides in the geometry and in the boundary conditions
- Each condition is an **assertion about the part of the circulation that was not modelled**

Inlet I · The flow waveform



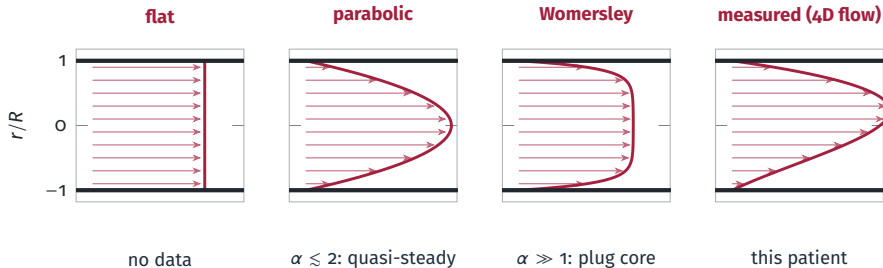
Inflow waveforms of two patients from 4D-flow MRI, and a waveform with 25% reduced stroke volume; right, measured inlet velocity profiles at five phases.

Sources of $Q(t)$, in decreasing order of fidelity:

- 1 Phase-contrast or 4D-flow MRI through the inlet plane, same patient
- 2 Doppler velocity with an assumed cross-sectional area
- 3 A literature waveform scaled to the cardiac output

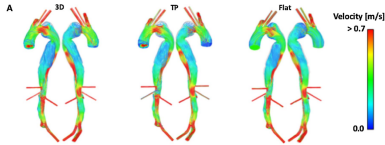
Stroke volume and waveform shape matter: a 25% lower stroke volume reduced peak velocity by 28% and TAWSS by up to 35%; another patient's waveform altered predicted pressure by up to 25%.

Inlet II · The velocity profile, chosen from α and Re

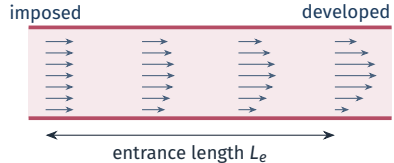


- Only the flow rate is measured by Doppler; its distribution over the section must be assumed. With $\alpha \approx 20$ the aortic profile is plug-like: **Womersley, not parabolic**
- 4D-flow MRI measures the profile itself (three components on the inlet plane), including the skew and eccentricity produced by the valve

How far downstream does the imposed profile matter?



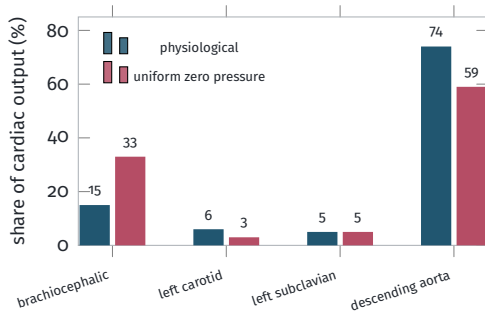
Peak systolic streamlines with a three-directional (3D), through-plane (TP) and flat inlet profile.



- Viscous diffusion progressively erases the imposed profile; the distance depends on Re and α
- Differences between profiles are confined to the ascending aorta and arch: TAWSS within 6%, pressure within 1%

Near the inlet the profile is a first-order choice; beyond the entrance length it is second-order. Place the inlet far from the lesion, or measure the profile.

Outlets I · There is no neutral outlet condition



- Each outlet stands for an entire downstream vascular bed
- Imposing the same pressure at every outlet divides the flow according to the resistance of the **modelled stubs**, i.e. according to where the truncation planes were placed
- Here the descending aorta, and hence the coarctation, would receive 20% less flow than it should

The flow through the coarctation, and therefore Δp , is set by the **outlet conditions**. Declining to choose is itself a choice.

Outlets II · Lumped-parameter outlet models

prescribed pressure



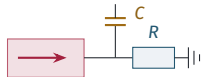
no downstream bed;
flow split by the stubs

resistance



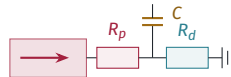
$p = RQ$: correct mean split;
pressure in phase with flow

two-element Windkessel



adds compliance:
diastolic decay $\tau = RC$

three-element (RCR)



$R_p \approx Z_c$ absorbs
the incident wave

increasing physiological fidelity; increasing number of parameters to justify

Each outlet carries a zero-dimensional model of the bed that was truncated: the lumped models of lecture 2, reused as boundary conditions.

Outlets III · Calibrating the three-element Windkessel

BOARD WORK

per outlet i , from measured $\bar{Q}_i$ (4D flow) and $\bar{p}$ (cuff):

$$R_{p,i} + R_{d,i} = \bar{p} / \bar{Q}_i \quad (\text{mean pressure})$$

$$R_{p,i} \approx Z_{c,i} = \rho c / A_i \quad (\text{no reflection; } c = \text{pulse wave velocity})$$

$$C_i = \tau / R_{d,i} \quad (\tau = \text{diastolic decay time, lecture 2})$$

- $R_p + R_d$: mean pressure; C: pulse pressure;
 R_p : wave absorption; then simulate,
compare, iterate

outlet	R_p $\text{g mm}^{-4} \text{ s}^{-1}$	C $\text{mm}^4 \text{ s}^2 \text{ g}^{-1}$	R_d $\text{g mm}^{-4} \text{ s}^{-1}$
brachiocephalic	0.0294	12.19	0.1281
left carotid	0.0872	1.30	1.3955
left subclavian	0.0418	3.77	0.4682
descending aorta	0.0203	10.02	0.1583

*parameters of the reference model: resistances scale
inversely with the flow share*

Every parameter traces to a measurement or to a stated assumption (here: 80 mmHg mean pressure at all outlets, since no invasive data were available).

Wall · Rigid or compliant?

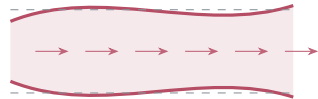
rigid wall



no-slip velocity, no motion. The standard assumption: over-estimates peak WSS by 10–30%, removes wave propagation; a **bounded, characterised** error

- Reference model: FSI, $E = 878$ kPa, thickness 1 mm from the literature (pulse wave velocity ≈ 5 m/s)
- Numerically delicate: wall and blood have similar density; loosely coupled schemes are unstable

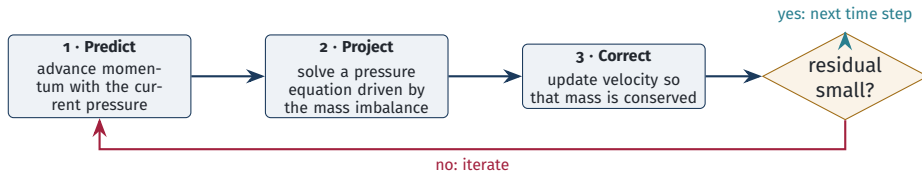
fluid–structure interaction



the wall deforms under pressure and feeds back on the flow. Several times the cost; requires **stiffness and thickness**, rarely known for the patient

A compliant wall combined with a hand-thresholded lumen misallocates effort: the wall error is smaller than the segmentation error.

Step 5 · The solver: a pressure-correction scheme



BOARD WORK

1. predictor, from $\mathbf{u}^n, p^n$: $\rho \frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = -\rho(\mathbf{u}^n \cdot \nabla)\mathbf{u}^n - \nabla p^n + \mu \nabla^2 \mathbf{u}^n$ (satisfies momentum, not mass)
2. pressure correction p' : $\nabla^2 p' = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*$ (the mass imbalance is the source term)
3. correction: $\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p'$, $p^{n+1} = p^n + p' \Rightarrow \nabla \cdot \mathbf{u}^{n+1} = 0$

A pressure equation is manufactured from the continuity constraint; the solution is approached iteratively.

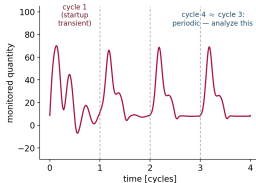
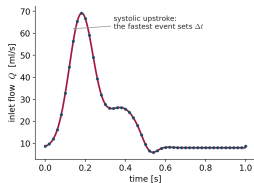
Steady versus transient simulation

steady $\partial/\partial t = 0$

one set of boundary values (mean or peak flow); the iteration loop is run once, to convergence; minutes to hours

transient time-marching with step Δt

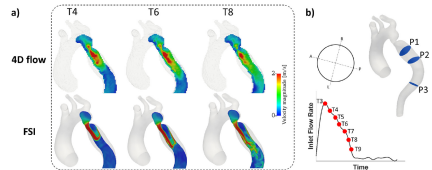
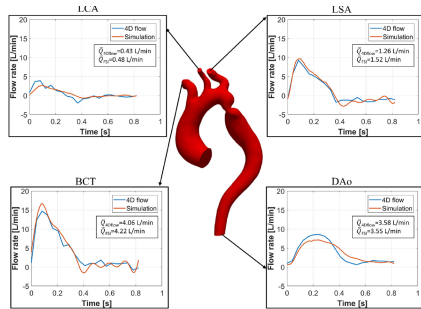
boundary values change every step; the loop is run at each of $\approx 10^3$ steps per cycle, over several cycles; hours to days



resolve the waveform: ≥ 500 – 1000 steps/cycle ($\Delta t \approx 1$ – 2 ms), CFL $= u\Delta t/\Delta x \lesssim 1$

- Time step from the Courant condition $u \Delta t/\Delta x \lesssim 1$, binding in the jet: $\Delta t \approx 1$ – 2 ms
- The first cycles carry the start-up transient and are discarded; the reported cycle must reproduce the previous one (reference: 5% criterion)

Step 6 · Validation against measurement

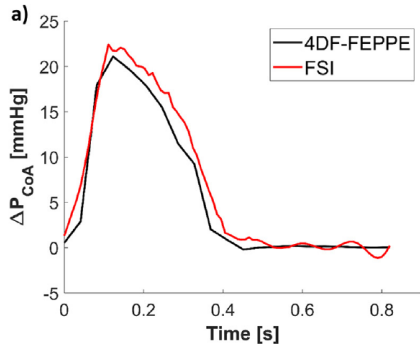


velocity magnitude in the descending aorta at three systolic phases

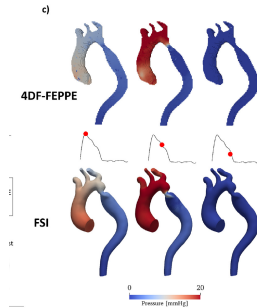
outlet flow rates: simulation (red) versus 4D-flow MRI (blue)

- Outlet flows: correlation 0.88–0.98; velocity contours: normalised RMSE 0.17–0.36
- **Verification:** are the equations solved correctly? **Validation:** are the correct equations solved?

The answer: the trans-coarctation pressure drop



4D



- Peak-to-peak drop 16.9 mmHg (simulation) against 17.6 mmHg from the measured velocities; bias below 1 mmHg along the aorta; both below the 20 mmHg threshold
- The same physics can also be applied *directly* to the measured velocity field: velocity in, pressure out, without a full simulation

Part 1 in one table

step	choice in the coarctation model	what it inherits
clinical question	peak-to-peak Δp ; threshold 20 mmHg	the decision
geometry	MR angiography, manual segmentation	$\Delta p \propto R^{-4}$: dominant uncertainty
domain	inlet at the ascending aorta; arch branches and descending aorta retained	omitted branches redistribute flow
physics	Newtonian, laminar; Re , α evaluated first	no closure; Stokes layer 1 mm
mesh	4 M tetrahedra, 0.2 mm boundary layer; GCI 3.4%	WSS from the first cell
inlet	measured waveform, parabolic profile 4.5 D upstream	profile erased with distance
outlets	three-element Windkessel, calibrated on measured flows	sets the flow through the lesion
wall	FSI, literature properties	wave propagation; assumed stiffness
solver	transient, $\Delta t = 2$ ms, 2 cycles, residual 10^{-5}	convergence verified
validation	outlet flows and velocity fields vs 4D flow	the loop closed

Break · 10 minutes

Part 2: the same workflow, applied to coronary stenosis and FFR-CT.

Part 2 · Coronary stenosis and FFR-CT

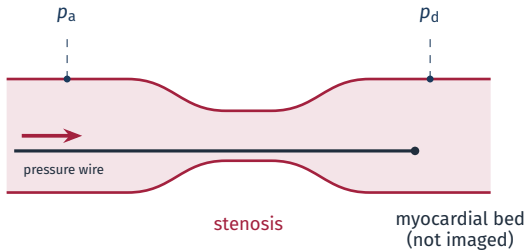
the condition · the invasive criterion and its limits · the same workflow: what changes when the vessel is 3 mm wide and embedded in a contracting muscle

Coronary artery disease: the condition

- Atherosclerotic plaque narrows the epicardial arteries (3–4 mm proximally, < 1.5 mm distally); the leading cause of death worldwide
- Ischaemia occurs when a stenosis limits flow to the myocardium under demand; angina, infarction
- Treatment: medical therapy, percutaneous intervention (stenting), bypass surgery

Anatomical severity predicts **functional** significance poorly; revascularisation should rest on function.

The diagnostic criterion: invasive fractional flow reserve



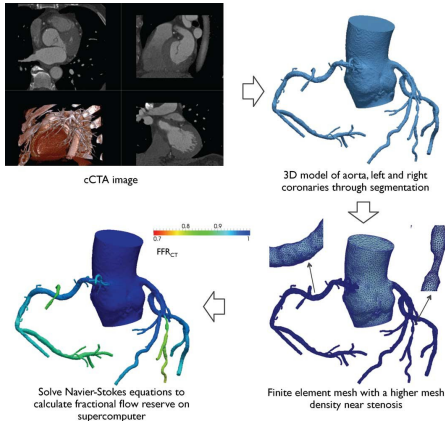
BOARD WORK

$$\text{FFR} = \frac{\bar{p}_d}{\bar{p}_a} \text{ at maximal hyperaemia;}$$

revascularise if $\text{FFR} \leq \mathbf{0.80}$

- A pressure wire is advanced past the lesion during catheterisation; adenosine induces maximal vasodilation of the microcirculation
- FFR is the fraction of the normal maximal flow that the lesion still allows; 1.0 means no obstruction
- Limitations: invasive, adenosine (discomfort, contraindications), time and cost; used in a minority of eligible patients

Computing FFR *in silico*: the FFR-CT workflow

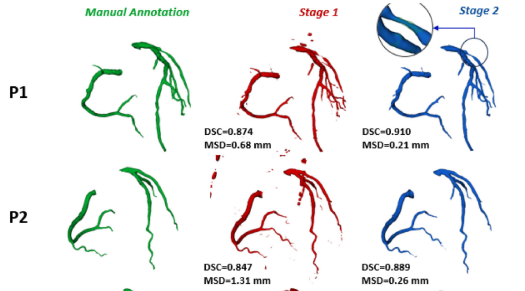


From coronary CT angiography to a map of FFR-CT along the tree.

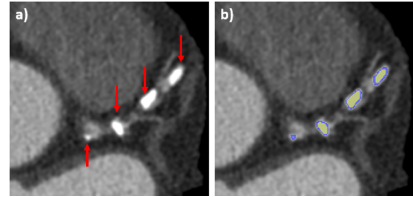
- The same six steps as in Part 1: image, geometry, mesh, boundary conditions, solver, validation
- Output: the pressure field under simulated hyperaemia, hence FFR at every point of the tree
- FFR-CT is a regulated, reimbursed diagnostic: CFD operating as a medical device

Same equations, same solver. What changes: the imaging, the domain, and above all the **outlet conditions**, because the downstream bed is now a contracting muscle in a pharmacological state.

Step 2 · Imaging and segmentation of the coronary tree



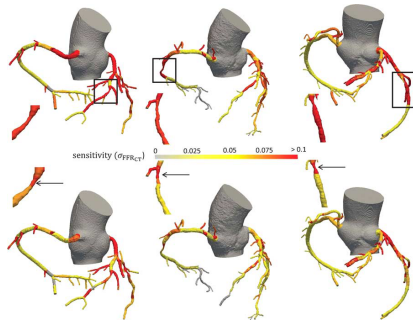
Manual annotation (green) and two-stage neural-network segmentation (red, blue) of the coronary tree; Dice 0.895, mean surface distance 0.47 mm.



Calcified plaque on CT angiography and its segmentation

- Coronary CT angiography: 0.4 mm voxels for a 3 mm vessel; ECG-gated, contrast-enhanced
- Calcium **blooms** and narrows the apparent lumen; motion and stents degrade the image
- Automated segmentation removes operator bias, not the ambiguity of the boundary

Sensitivity of FFR-CT to the lumen boundary



Standard deviation of FFR-CT induced by uncertainty in the segmented lumen; the largest sensitivity sits at the lesions.

BOARD WORK

resistance of a stenosis $\propto R^{-4}$; R under-read by 10% $\rightarrow$ +52% in Δp

- The R^{-4} sensitivity of Part 1, now on a 1.5 mm radius: one voxel is 25% of the radius
- The minimal lumen area dominates the uncertainty of FFR-CT near the 0.80 threshold

As in Part 1, the lumen boundary is the first-order determinant of accuracy; here the margin for error is a fraction of a millimetre.

Modelling choices for a coronary model

domain

aortic root plus both coronary trees, truncated below ≈ 1.5 mm; each truncation is an outlet with its own bed

rheology

shear rates of $100\text{--}700\text{ s}^{-1}$: Newtonian is adequate; shear-thinning matters only behind a severe stenosis

BOARD WORK

coronary: $D \approx 3$ mm, $U \approx 0.2$ m/s, 72 bpm

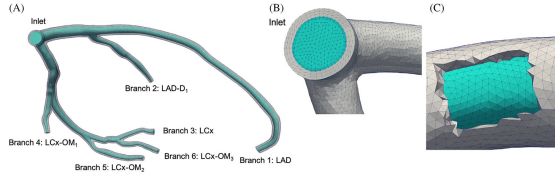
$$\text{Re} = \frac{\rho U D}{\mu} \approx 180 \quad (\text{hyperaemia: } \approx 500)$$

$$\alpha = R\sqrt{\omega\rho/\mu} \approx 2.3 \quad \delta \approx 0.9\text{ mm} \approx 0.6 R$$

- $\text{Re} \ll 2300$: laminar, even under hyperaemia
- $\alpha \approx 2$: quasi-steady, near-parabolic profiles; the Stokes layer spans most of the radius

Small α : the flow at each instant is close to the steady flow at that instant, which makes a **steady hyperaemic simulation** defensible for FFR.

Step 3 · Meshing a coronary tree

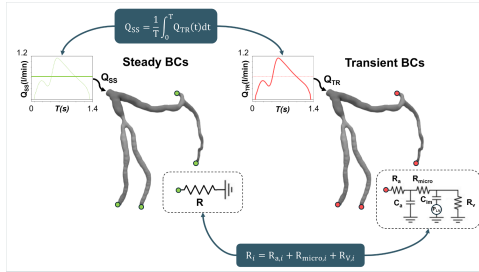


Segmented left coronary tree, and the mesh with prismatic boundary layers at the inlet section and at the wall.

- Elements of 0.1–0.2 mm in a 3 mm vessel; 1–3 million elements for a full tree
- Boundary layers again: with $\delta \approx 0.9$ mm and $R = 1.5$ mm the near-wall gradient occupies most of the section
- Refinement at the stenosis, where velocity and pressure gradients are steepest and the answer is read

Mesh independence assessed on FFR at the lesion, not on a global velocity.

Step 4 · Boundary conditions: inlet and the myocardial bed

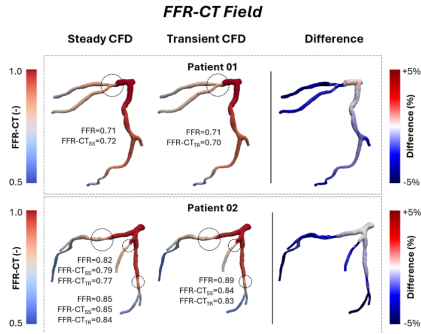


Steady (mean-flow, resistance outlets) versus transient (waveform, lumped coronary outlets with intramyocardial pressure) boundary conditions for a coronary model.

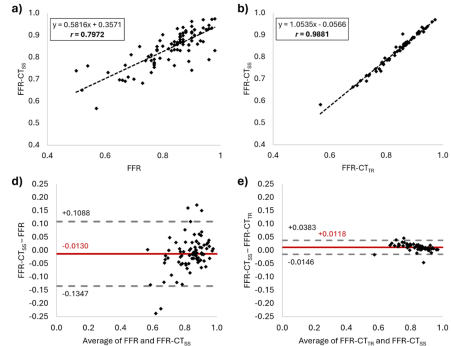
- **Inlet:** aortic pressure from the brachial cuff, or the flow of the aortic root; no patient-specific coronary flow is available non-invasively
- **Outlets:** the resistance of each myocardial territory is not measurable; it is allocated from the **myocardial mass** (allometric scaling) and distributed among branches by their diameter (Murray's law, $Q \propto D^{2.7}$)
- **Hyperaemia** is simulated by reducing the microvascular resistance to ≈ 0.24 of its resting value

Coronary flow is predominantly **diastolic**: the myocardium compresses the intramural vessels. A transient model needs an intramyocardial pressure in the outlet; a steady hyperaemic model needs only the resistances.

Steady or transient? What the comparison shows



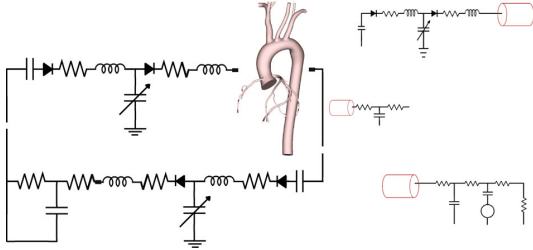
FFR-CT along two left coronary trees, steady versus transient, and their pointwise difference



133 vessels: steady FFR-CT against invasive FFR (a, d) and against transient FFR-CT (b, e)

- Steady versus transient: $r = 0.99$, limits of agreement ± 0.03 , pointwise FFR-CT within 2%, at 1/30 of the computing time
- Either against invasive FFR: $r \approx 0.8$, limits of agreement ± 0.13 : the model-to-measurement gap is four times larger

Closing the loop: the heart as a boundary condition; wall and solver

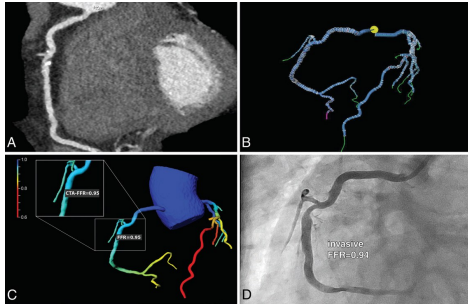


Multiscale coronary model: three-dimensional aorta and coronaries coupled to a lumped heart and to lumped coronary beds with intramyocardial pressure.

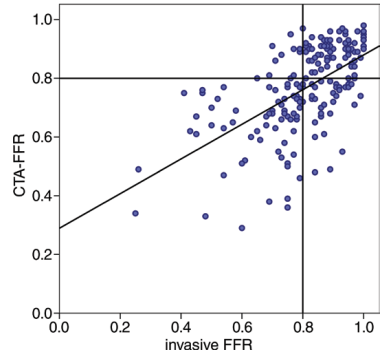
- When aortic pressure is not imposed, a lumped heart model (lecture 2) supplies it: the three-dimensional domain becomes one element of a closed loop
- **Wall:** rigid in all clinical FFR-CT implementations; motion of the heart is neglected
- **Solver:** unchanged, the pressure-correction scheme of Part 1; steady hyperaemic solution for FFR, transient where the waveform is of interest

The multiscale principle of lecture 2: the Windkessel and the heart model are not replaced by the 3-D simulation; they become its boundaries.

Step 6 · Validation against invasive FFR



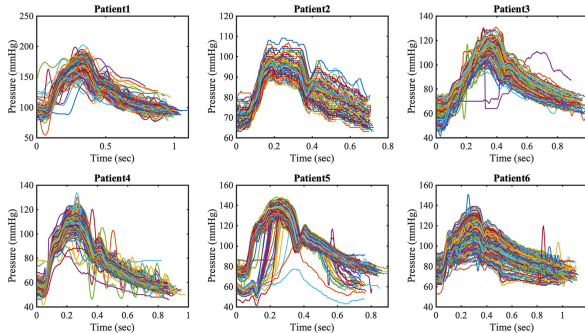
One patient: CT angiography (A), segmented tree (B), FFR-CT map (C) and invasive angiography with wire FFR 0.94 (D).



FFR-CT versus invasive FFR per vessel; the lines mark the 0.80 threshold

- Diagnostic accuracy of $\approx 75\text{--}85\%$ against the invasive standard; the scatter around the threshold is where decisions are uncertain
- Validation against the **clinical** reference in a cohort, not against a velocity field in one patient

Where the uncertainty comes from

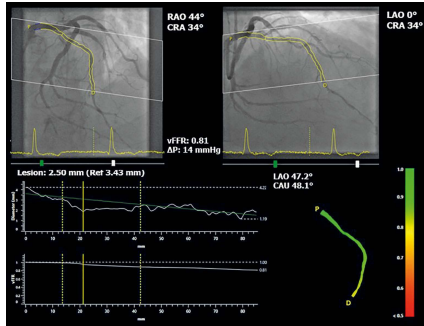


Aortic pressure waveforms for six patients, sampled from the uncertainty of the clinical inputs, used to propagate input uncertainty through coronary simulations.

- Inputs: lumen segmentation, myocardial mass, cuff pressure, assumed hyperaemic response
- Models: the resistance allocation and the rigid wall
- Numerics: the smallest term, once a mesh study has been done

The ranking of Part 1 holds: geometry and outlet parameters first, constitutive models second, numerics last. Report the spread, not the value.

A variant: FFR from invasive angiography



Three-dimensional reconstruction of a coronary segment from two angiographic projections, and the computed pressure drop and vessel-FFR along it.

- Same workflow, different image: the geometry is reconstructed from two X-ray projections acquired during catheterisation
- Aortic pressure is measured at the catheter tip; no pressure wire and no adenosine
- Result in minutes, in the catheterisation laboratory

The image is interchangeable; the physics, the boundary-condition problem and the uncertainty budget are not.

Two diseases, one workflow

step	aortic coarctation	coronary stenosis (FFR-CT)
clinical question	peak-to-peak $\Delta p > 20$ mmHg	FFR ≤ 0.80 at hyperaemia
invasive reference	catheter pull-back	pressure wire and adenosine
imaging	MR or CT angiography; 4D-flow MRI for flow	coronary CT angiography; no flow data
domain	arch with three branches; $D \approx 25$ mm	aortic root and two trees; $D \approx 3$ mm
regime	Re up to 6000, $\alpha \approx 20$: transitional, plug-like	Re ≈ 200 –500, $\alpha \approx 2$: laminar, quasi-steady
inlet	measured waveform; profile from α	cuff pressure or aortic flow
outlets	Windkessel calibrated on measured flows	resistances allocated from myocardial mass and Murray's law; hyperaemia imposed
wall	rigid or FSI	rigid
simulation	transient, several cycles	steady hyperaemic (transient optional)
validation	flow fields vs 4D flow, one patient	invasive FFR, cohorts
dominant uncertainty	lumen; outlet parameters	lumen; microvascular resistance

Summary · three central results

1 · The question determines the model

a pressure drop or a pressure ratio is inferred from geometry and boundary data through the conservation laws, and inherits every assumption in both

2 · Re and α precede every choice

they fix the flow regime, the inlet profile, the near-wall mesh, and whether a steady simulation is admissible

3 · Lumen and outlets dominate the uncertainty

10% in radius is 50% in Δp ; the unmeasured downstream bed is the second term; numerics come last

The practice that outlasts all three: **report the spread, not the value.**

Reference values

Quantity	Value	Consequence
coarctation threshold	peak-to-peak Δp 20 mmHg	indication for repair
FFR threshold	0.80	indication for revascularisation
radius sensitivity	$\Delta p \propto R^{-4}$, $\tau_w \propto R^{-3}$	10% in radius $\rightarrow$ 52% in Δp
ascending aorta	Re 1700 (mean), 5500 (peak); $\alpha \approx 20$	transitional; plug-like profile; Stokes layer 1 mm
coronary artery	Re 200–500; $\alpha \approx 2$	laminar; quasi-steady; near-parabolic
first near-wall cell	$\lesssim 0.1 \delta$	WSS error below 5%
physiological WSS	0.5–2 Pa	transmural pressure $\approx$ 13 kPa

Sources of the figures, and further reading

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